

PCEP – Python Cheat Sheet (Exam-Focused)

1) Syntax & Basics

- Indentation creates blocks (PEP 8 recommends 4 spaces).
- Comments: # line, triple quotes for docstrings.
- I/O: input(), print(x, y, sep=" ", end="").
- Types: int, float, str, bool; type casting via int(), float(), str().

2) Operators

- Arithmetic: + - * / // % **
- Comparison: == != < <= > >=
- Logical: and, or, not
- Membership: in, not in
- Assignment: =, +=, -=, *=, /=, //=, %=, **=

3) Control Flow

if cond: ... elif cond: ... else: ...

while cond: ... (break, continue, else)

for item in iterable: ... (break, continue, else)

range(stop) / range(start, stop[, step])

4) Collections

- List: [], mutable; common ops:
append(x), insert(i,x), extend(it), pop([i]), remove(x), index(x), count(x), sort(), reverse()
slicing: a[i:j:k]; len(a); min/max/sum.
- Tuple: (), immutable; packing/unpacking.
- Dict: {k:v}, mutable; d[k], get(k, default), keys(), values(), items(), pop(k), update({...})
- String: immutable; methods: lower(), upper(), strip(), split(), join(), find(), replace()
slicing: s[i:j:k]; membership: ch in s

5) Functions

def name(params=defaults): return value

- Positional, keyword, default parameters; return ends the function.
- Scope: local vs global; use global x to rebind module variable (not recommended).

6) Exceptions

try: ... except SomeError: ... else: ... finally: ...

- Common: ZeroDivisionError, ValueError, TypeError, IndexError, KeyError.

7) Built-ins & Idioms

- len(), type(), int(), float(), str(), bool(), list(), tuple(), dict()
- enumerate(seq), range(), sorted(it), reversed(seq)
- Multiple assignment: a, b = b, a
- Truthiness: 0, 0.0, "", [], {}, None are Falsey.

8) Useful Patterns

- Input to int: n = int(input()) # may raise ValueError

- Loop with index: `for i, x in enumerate(xs): ...`
- Dict safe lookup: `d.get(key, default)`
- Guard division: `if b != 0: a/b else: handle`